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Pre-owned housing price index forecasts using Gaussian process regressions

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Abstract

Purpose – The purpose of this study is to make property price forecasts for the Chinese housing market that has grown rapidly in the last 10 years, which is an important concern for both government and investors.

Design/methodology/approach – This study examines Gaussian process regressions with different kernels and basis functions for monthly pre-owned housing price index estimates for ten major Chinese cities from March 2012 to May 2020. The authors do this by using Bayesian optimizations and cross-validation.

Findings – The ten price indices from June 2019 to May 2020 are accurately predicted out-of-sample by the established models, which have relative root mean square errors ranging from 0.0458% to 0.3035% and correlation coefficients ranging from 93.9160% to 99.9653%.

Originality/value – The results might be applied separately or in conjunction with other forecasts to develop hypotheses regarding the patterns in the pre-owned residential real estate price index and conduct further policy research.

Keywords Pre-owned housing price index, Price forecasting, Gaussian process regression, Bayesian optimization, Cross-validation, Chinese real estate

Paper type Research paper

1. Introduction

The Chinese real estate market has grown significantly during the past 10 years. Real estate price prediction problems have surely become one of the biggest issues facing governments and investors (Xu and Zhang, 2022d; Xu and Zhang, 2024a; Tekouabou *et al.*, 2023; Xu and Zhang, 2022a; Lahmiri *et al.*, 2023; Xu and Zhang, 2023m). It is critical to comprehend real estate price trends and changes because they have a direct impact on people's decisions about where to live, how to invest in real estate and how regulatory authorities formulate and implement their regulations (Xu and Zhang, 2023a; Hjort *et al.*, 2022; Xu and Zhang, 2023b; Yang *et al.*, 2023; Xu and Zhang, 2022c). Predicting real estate prices is of course of interest to many customers and suppliers of estimates.

Predicting financial and economic time-series data with high accuracy and dependability is of interest to a large number of scholars and practitioners. Many of their expansions and modifications as well as certain important time series approaches, such the autoregressive (AR), vector autoregressive (VAR) and vector error correction (VEC) models, have been studied

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for a range of forecasting applications (Xu and Zhang, 2023c; Xu and Zhang, 2023d; Kim *et al.*, 2007; Xu and Zhang, 2023e; Xu and Zhang, 2023f; Cabrera *et al.*, 2011; Xu, 2020; Xu, 2019a; Webb *et al.*, 2016; Xu, 2017a; Xu, 2017b; Wei and Cao, 2017; Xu, 2018a; Xu, 2017; Yang *et al.*, 2018; Xu, 2018b; Xu and Zhang, 2021a; Liu and Wu, 2020; Xu and Zhang, 2024b; Xu and Zhang, 2023g; Milunovich, 2020; Xu and Zhang, 2022b; Xu and Zhang, 2021b; Glennon *et al.*, 2018; Xu, 2019b; Xu, 2018c; Guo, 2020; Xu, 2018d; Xu, 2018e; Mei and Fang, 2017; Xu, 2015; Xu, 2019; Hepşen and Vatansever, 2011; Xu and Zhang, 2023h; Baroni *et al.*, 2005). Recently, it has been discovered that many machine learning techniques, such as neural networks, boosting, bagging, regression trees (RT), random forests, nearest neighbors, deep learning, ensemble learning and Gaussian process regressions, are promising and effective solutions to a range of real estate price forecasting problems. The neural network model seems to be one of the most popular ways (Xu and Zhang, 2024c; Xu and Zhang, 2021c; Xu and Zhang, 2023i; Xu and Zhang, 2023j; Xu and Zhang, 2023k) for predicting real estate prices (Liu and Wu, 2020; Milunovich, 2020; Lam *et al.*, 2008; Yasnitsky *et al.*, 2021; Xu and Zhang, 2023 1), and these evaluations, while not comprehensive, are generally consistent with a range of empirical studies on the adoption of machine learning techniques for forecasting in finance and economics (Xu and Zhang, 2022f; Xu and Zhang, 2023n; Yang *et al.*, 2008; Xu and Zhang, 2024d; Xu and Zhang, 2022g; Wang and Yang, 2010; Xu and Zhang, 2021d; Xu and Zhang, 2023o; Yang *et al.*, 2010; Xu and Zhang, 2022h; Xu and Zhang, 2023p; Wegener *et al.*, 2016; Xu and Zhang, 2023q; Xu and Zhang, 2022i; Xu and Zhang, 2023r; Xu and Zhang, 2022e). On the other hand, the predictions of time-series data from the real estate price index using the Gaussian process regression are not well studied.

A novel regression method is based on Neal's work on Bayesian learning for neural networks (Neal, 2012). Modeling noisy data is appropriate because the method relies on priors over functions of Gaussian processes. It was demonstrated that a wide variety of neural network-based Bayesian regression models will converge to Gaussian processes at the limit of an infinite network (Neal, 2012). Regressions using the Gaussian process have demonstrated efficacy in simulating both noisy (Jin and Xu, 2024a; Jin and Xu, 2024b) and noise-free (Neal, 1997) data. Brahim-Belhouari and Vesin (2001) examined Gaussian processes with radial basis function neural networks for forecasting problems requiring stationary time-series data and discovered that Bayesian learning yields superior prediction results. According to research by Brahim-Belhouari and Bermak (2004), it is beneficial to look at different covariance functions, which is the method used in this study. In addition, prediction techniques based on Gaussian processes can be used to effectively solve forecasting problems for nonstationary time-series data. Brahim-Belhouari and Bermak's (2004) research shows that Gaussian process regressions outperform radial basis function neural networks. Furthermore, the Gaussian process formulation's value and benefit come from the exact matrix operations required to integrate the prior and noise models (Brahim-Belhouari and Bermak, 2004). Brahim-Belhouari and Bermak (2004) also recommended using the technique of multimodel forecasting using Gaussian process predictors, which is comparable to how we would use model averaging. Based on a number of recent research (Xu and Zhang, 2023s; Jin and Xu, 2024c), price time series for the resource and agricultural industries may be successfully projected using Gaussian process regressions.

Forecasting efforts for residential real estate prices have also been noticed using traditional econometric models and, more recently, machine learning approaches. In comparing parametric and semiparametric classical econometric models, for instance, Gençay and Yang (1996) and Gencay and Yang (1996) found that semiparametric models are better suited for predicting and assessing residential housing prices. Glennon *et al.* (2018) find that the accuracy of property appraisals based on house price indices is increased when many models are used. By devising a system to mitigate the effects of measurement errors

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associated with the repeat sales methodology, Clapp and Giaccotto (1992) demonstrate the superior efficacy of the assessed value strategy. Forecasts produced using formulas based on disaggregated data for the entire city have an edge over forecasts created using local average pricing formulas, claim Kaboudan and Sarkar (2007). Mei and Fang (2017) use trend analysis and multiple regression to build a dynamic state forecasting model of the average selling price. Levesque (1994) breaks down residential property prices using a case study of airport noise. Hepşen and Vatansever (2011) analyze the AR integrated moving average model's performance. Baroni *et al.* (2005) use principal components analysis to generate an indicator of repeat sales that forecasts apartment prices. Guo (2020) uses regression models, both linear and nonlinear, to study future price stability. Paris (2008) looks into artificial neural networks for machine learning techniques in order to forecast changes in local and national price indices for the UK residential real estate market. Chi (2017) proposes estimating residential real estate prices via a spatial back-propagation neural network. Bee-Hua (2000) suggests merging neural networks and evolutionary algorithms to assess demand for residential building. When it comes to calculating the values of residential real estate, Štubňová *et al.* (2020) find that neural networks perform better than regression models. Seya and Shiroi (2022) examine the closest neighbor Gaussian process and deep neural network for residential unit rent price prediction and find that the former has more promise. Rafiei and Adeli (2018) recommend using an unsupervised deep Boltzmann machine (DBM) learning approach, a softmax layer to extract relevant features from the input data and a three-layer back-propagation neural network (or support vector machine) to convert the trained unsupervised DBM into a supervised regression network to estimate construction cost from economic variables and indices. Zhang *et al.* (2021) found that the radial basis function-based support vector regression (SVR) algorithm and the extra-trees regression approach are both useful for simulating the fine-scale spatiotemporal distribution of residential land values. Yoo *et al.* (2012) use cubist, random forest and traditional ordinary least squares to simulate the hedonic pricing of residential real estate sales. They find that the random forest model yields the best accurate results. Researchers Dimopoulos and Bakas (2019), Hong *et al.* (2020) and Dimopoulos *et al.* (2018) demonstrate how machine learning models may be applied to improve the precision of real estate mass evaluations. Ai *et al.* (2020) suggest that machine learning algorithms are useful for residential land appraisals as well. Picchetti (2017) shows how sample variation in hedonic geospatial residential property price evaluations may be overcome by applying the gradient tree boosting technique.

1.1 Research contributions and limitations

To keep with this topic, we concentrate on Gaussian process regressions for estimates of pre-owned residential real estate price indices for ten major Chinese cities from March 2012 to May 2020, a time when the real estate market expanded rapidly. Results here contribute to the literature in the following aspects. First, to the best of our knowledge, this is the first forecast research that looks at pre-owned residential real estate price indices in the Chinese market using Gaussian process regressions. In earlier studies, the Boston housing-related problems were often examined from the perspective of property valuation using the Gaussian process regression technique (Harrison and Rubinfeld, 1978; Rasmussen, 1997; Williams, 1997; Chu *et al.*, 2004; Kim and Ghahramani, 2006; Titsias, 2009; Tokdar *et al.*, 2010; Jylänki *et al.*, 2011; Nielsen *et al.*, 2012; Friedman and Nachman, 2013; Muñoz-González *et al.*, 2014; Dearmon and Smith, 2016; Ranjan *et al.*, 2016; Hartmann and Vanhatalo, 2019; Li *et al.*, 2021; Miao *et al.*, 2021; Murakami, 2021; Miao *et al.*, 2022; Sebenius *et al.*, 2022; Khosravi *et al.*, 2022; Liu *et al.*, 2022; Algikar and Mili, 2023). Second, it might not take much

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incentive given how well-known the Chinese residential real estate sector is. For investors and policymakers, forecasting residential real estate price indices should be both essential and possibly challenging, because having a solid understanding of pricing patterns may aid in making decisions. Numerous forecasting techniques exist, such as machine learning and econometric models. The nonlinear patterns displayed by the residential real estate price indices under examination, along with the literature's acknowledged worth and potential for real estate price forecasting, led to the selection of the Gaussian process regression approach, which could effectively model nonlinear characteristics embedding the underlying price indices. Third, a variety of basis functions and kernels will be used in a Bayesian optimization method together with the cross-validation technique to train the Gaussian process regression. Particularly, Bayesian optimizations have great potential to improve the forecast models' generalization performance and flexibility. Fourth, we believe that our results will contribute to a better understanding of using machine learning technology to anticipate residential real estate price in China, because the majority of previous studies have evaluated other forms of real estate assets based on a single location (Lam *et al.*, 2008; Ma *et al.*, 2015; Li *et al.*, 2018; Wang *et al.*, 2014; Ho *et al.*, 2021). Because supply and demand are most active in these major cities and each may have unique price characteristics that merit examination, the current work's coverage of these cities along with the data's accessibility should represent an economically natural way to investigate the residential real estate forecast problem. Fifth, our findings may offer a more modern viewpoint on the effectiveness of machine learning techniques for real estate price index estimates for the Chinese market, as previous research in this field has generally focused on earlier time periods when researching other forms of real estate, such as Q12000–Q32010 (Wang *et al.*, 2014), 7M2013–12M2013 (Ma *et al.*, 2015), 6M1996–8M2014 (Ho *et al.*, 2021), 12M2010–10M2017 (Liu and Liu, 2019) and 1M2005–11M2018 (Li *et al.*, 2020). The level of uncertainty that currently exists in the residential real estate market might lead to more nonlinear pricing behavior (Xu and Zhang, 2024e; Xu and Zhang, 2022j). By developing Gaussian process regression models based on a more current time period, from March 2012 to May 2020, this work contributes to the body of literature on the applicability of the Gaussian process regression for real estate price index projections in dynamic situations. We also apply the developed models on pre-owned housing prices across the ten cities investigated during a more recent time period of February 2023–January 2024, and find at the models could still generate rather accurate forecast performance for November 2023, December 2023 and January 2024. Sixth, this study offers timely forecasting tools that policymakers and investors may find useful. Machine learning techniques could seem more complicated than econometric models to many forecast consumers who are not as experienced. Therefore, we create very simple yet accurate Gaussian process regressions to support technological predictions. Since econometric and machine learning models may overfit or underfit, we have considered a trade-off between stabilities and prediction accuracy in our model building processes. In particular, we perform out-of-sample forecasts from June 2019 to May 2020 and get relative root mean square errors (RRMSE) ranging from 0.0458% to 0.3035% and correlation coefficients ranging from 93.9160% to 99.9653% for the ten price indices. We also conduct benchmark analysis of the constructed models here against different traditional time-series econometric models and alternative machine learning models, and find that the Gaussian process regression models built here lead to statistically significant better performance in terms of forecast accuracy. Our results might be applied separately or in conjunction with other projections to develop hypotheses regarding the patterns in the residential real estate price index and conduct further policy research. It should be meaningful to note that the current technique adopted here relies on

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historical data as the source of prediction information rather than considering other economic aspects as additional potential predictors. As a consequence, one may integrate the prediction results from other sources with those generated by the models built here to overcome this potential limitation. In this situation, it could be necessary to build suitable model combinations.

2. Literature review

2.1 Real estate price forecasting using econometric methods

While machine learning approaches for residential real estate price index prediction are the focus of this study, econometric methodologies for broad real estate price forecasts have also been thoroughly investigated in the literature. Rental prices for the UK market are predicted using a variety of techniques, such as moving average, AR, integrated AR-moving average, simple least squares and seemingly unrelated regression equations (Silver and Goode, 1990; McGough and Tsolacos, 1995). In addition, the VAR regression approach indicates that the horizon affects how accurately retail rental prices are predicted for the UK market (Brooks and Tsolacos, 2000). Another study based on several linear least-squares regressions suggests that retail leasing costs in many British towns and cities may be predicted based on the size of the retail core and the demographics of the local population (Jackson, 2001). Time series analysis of retail real estate price returns has also been done. A generalized AR conditional heteroscedasticity in mean model indicates that macroeconomic factors are better predictors of price returns for office real estate in Australia than for retail or industrial real estate (West and Worthington, 2006). The VEC model suggests that retail sales and mortgage loans might be used to predict the price of Greek retail real estate. Numerous studies indicate that semiparametric models outperform parametric models in predicting residential real estate values (Gençay and Yang, 1996; Gencay and Yang, 1996). In addition, specific model combinations (Glennon *et al.*, 2018), such as combining nonlinear and linear models (Guo, 2020), may improve forecasting accuracy. It is also a good idea to combine trend analysis with different linear regression techniques to build dynamic models that predict average selling prices (Mei and Fang, 2017). Principal component analysis has been demonstrated to enhance residential real estate price projections (Baroni *et al.*, 2005); however, the AR integrated moving average approach also shows promise (Hepşen and Vatansever, 2011). In general, it is discovered that the VAR method and the directed acyclic graph technique are helpful for forecasting and studying the dynamics of Real Estate Investment Trusts in the USA (Kim *et al.*, 2007). Another study emphasizes how crucial it is to combine machine learning approaches, like neural networks and other nonparametric regression techniques, with econometric approaches, like the AR approach, the functional coefficient approach, and the exponential generalized AR conditional heteroscedasticity approach, to forecast returns of global securitized real estate (Cabrera *et al.*, 2011). Research has shown that machine learning models – most notably the neural network methodology – must be used in addition to econometric approaches for predicting real estate values (Kim *et al.*, 2007; Cabrera *et al.*, 2011; Liu and Wu, 2020). Several iterations of basic time-series econometric approaches are investigated in the context of real estate value forecasting research. For example, a generalized VAR technique that incorporates high-dimensional data processing approaches is used to analyze spillovers across residential home prices among 69 Chinese cities (Yang *et al.*, 2018); a jump generalized AR conditional heteroscedasticity model is used (Webb *et al.*, 2016); and the VEC technique incorporating the smooth transition weighting approach is used to improve the predictability of national housing price indices in the USA. Research has also suggested using a range of linear models to lessen the influence of nonlinearities on predictions of real estate prices

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(Webb *et al.*, 2016). A more direct approach may be to take into account nonlinear models, which include machine learning methods.

2.2 Real estate price forecasting using machine learning techniques

Geographically, the USA (Peterson and Flanagan, 2009; Plakandaras *et al.*, 2015), Australia (Milunovich, 2020), Europe (Ćetković *et al.*, 2018), Russia (Yasnitsky *et al.*, 2021), Nigeria (Abidoye and Chan, 2017; Abidoye and Chan, 2018), Uganda (Embaye *et al.*, 2021), Malaysia, China (Wei and Cao, 2017; Liu and Wu, 2020; Lam *et al.*, 2008; Ma *et al.*, 2015; Li *et al.*, 2018; Wang *et al.*, 2014; Ho *et al.*, 2021; Xu and Zhang, 2023t; Xu and Li, 2021), Italy (Morano *et al.*, 2015), Spain (Rico-Juan and de La Paz, 2021), Malawi (Embaye *et al.*, 2021), Tanzania (Embaye *et al.*, 2021), South Korea (Kang *et al.*, 2020) and Iran (Azadeh *et al.*, 2014; Rafiei and Adeli, 2016) are the countries and/or regions where machine learning techniques for real estate price projections have been applied. Given their widespread application, machine learning techniques hold promise for forecasting real estate prices using time-series data with a variety of underlying variables (Xu and Zhang, 2023u; Xu and Zhang, 2023v; Xu and Zhang, 2023w). The Chinese market is the main subject of our present investigation. Most of the material that we have looked at so far is focused on a few particular areas. For the USA, Fairfax County of Virginia, California, Boston and Ames, and Wake County of North Carolina (Peterson and Flanagan, 2009) are examined. For Malaysia, Petaling of Kuala Lumpur and Mukim Pulai of Johor Bahru are investigated. For China, Hong Kong (Lam *et al.*, 2008; Li *et al.*, 2018; Ho *et al.*, 2021), Tangshan (Gu *et al.*, 2011), Beijing (Li *et al.*, 2020; Yan and Zong, 2020), Chongqing (Wang *et al.*, 2014), Dalian, Shenzhen (Liu and Liu, 2019), Shanghai (Ma *et al.*, 2015), Xuzhou, Kunming, Handan, and Changchun (Liu and Wu, 2020) and Shenzhen, Beijing, Guangzhou, and Shanghai (Xu and Li, 2021) are explored. For Nigeria, Benin and Lagos (Abidoye and Chan, 2017; Abidoye and Chan, 2018) are assessed. For South Korea, Seoul is studied (Kang *et al.*, 2020). For Iran, Tehran is researched (Rafiei and Adeli, 2016). For Italy, Bari (Morano *et al.*, 2015), Taranto and Milan are taken into consideration. For Russia, Moscow is analyzed (Yasnitsky *et al.*, 2021). For Spain, Alicante is the focus (Rico-Juan and de La Paz, 2021). To provide a more thorough understanding of real estate price forecasts using machine learning techniques and to enrich the relevant literature with expanded empirical evidence on the usefulness of such methods in this research field, which could benefit both investors and policymakers in examining their applications to a wider selection of real estate markets, the current study takes into consideration the ten major cities in China, which represent the largest real estate markets in China [1].

Prior research has used modeling and forecasting for a range of purposes in developing machine learning systems, depending on their specialized fields. One of the areas of concentration may be developing machine learning algorithms based on various real estate attributes for asset appraisals (Peterson and Flanagan, 2009; Morano *et al.*, 2015; Abidoye and Chan, 2017; Abidoye and Chan, 2018; Kang *et al.*, 2020; Ho *et al.*, 2021; Xu and Li, 2021; Rafiei and Adeli, 2016), from many macroeconomic variables when valuing assets (Kang *et al.*, 2020; Rafiei and Adeli, 2016), from time series of lagging real estate prices for technical forecasts (Liu and Wu, 2020; Ma *et al.*, 2015; Wang *et al.*, 2014; Li *et al.*, 2020; Gu *et al.*, 2011), from various real estate attributes for technical forecasts (Lam *et al.*, 2008; Li *et al.*, 2018; Yasnitsky *et al.*, 2021; Liu and Liu, 2019; Yan and Zong, 2020; Embaye *et al.*, 2021; Rico-Juan and de La Paz, 2021; Chen *et al.*, 2017) and using several macroeconomic variables for technical forecasts (Wei and Cao, 2017; Milunovich, 2020; Lam *et al.*, 2008; Azadeh *et al.*, 2014; Ćetković *et al.*, 2018; Li *et al.*, 2018; Kang *et al.*, 2020; Yasnitsky *et al.*, 2021; Liu and Liu, 2019; Plakandaras *et al.*, 2015; Rico-Juan and de La Paz, 2021). Aside from the usual macroeconomic factors such as gross domestic product, interest rates and unemployment

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rates, other popular real estate characteristics taken into account include vintages, lot sizes and locations. Our work is one of several that creates machine learning models for technical forecasts using delayed time-series data on real estate prices (Liu and Wu, 2020; Ma *et al.*, 2015; Wang *et al.*, 2014; Li *et al.*, 2020; Gu *et al.*, 2011). To develop the models for this study, a number of kernels and basis functions are investigated through the use of Bayesian optimization and cross-validation. In particular, Gaussian process regressions are utilized.

Previous research has focused on single machine learning models (Lam *et al.*, 2008; Azadeh *et al.*, 2014; Ma *et al.*, 2015; Morano *et al.*, 2015; Abidoye and Chan, 2017; Ćetković *et al.*, 2018; Li *et al.*, 2018; Kang *et al.*, 2020; Yasnitsky *et al.*, 2021; Wang *et al.*, 2014; Li *et al.*, 2020; Gu *et al.*, 2011; Rafiei and Adeli, 2016; Chen *et al.*, 2017), comparisons between different models (Li *et al.*, 2017; Pai and Wang, 2020; Liu and Liu, 2019; Yan and Zong, 2020; Ho *et al.*, 2021; Embaye *et al.*, 2021; Xu and Li, 2021; Rico-Juan and de La Paz, 2021), machines learning models and conventional econometric models (Liu and Wu, 2020; Milunovich, 2020; Peterson and Flanagan, 2009; Li *et al.*, 2017; Abidoye and Chan, 2018; Plakandaras *et al.*, 2015) and combinations of models (Wei and Cao, 2017). More precisely, our findings indicate that the neural network approach is the most popular option for studies that concentrate on a single machine learning method (Lam *et al.*, 2008; Azadeh *et al.*, 2014; Ma *et al.*, 2015; Morano *et al.*, 2015; Abidoye and Chan, 2017; Ćetković *et al.*, 2018; Li *et al.*, 2018; Kang *et al.*, 2020; Yasnitsky *et al.*, 2021; Rafiei and Adeli, 2016), with SVRs (Wang *et al.*, 2014; Gu *et al.*, 2011; Chen *et al.*, 2017) and ensemble learners (Li *et al.*, 2020) coming in second and third, respectively. SVR models outperform back-propagation neural networks in terms of accuracy, according to a study evaluating many machine learning techniques. In another study, SVRs based on integrated hybrid genetic techniques achieve higher accuracy than fuzzy and back-propagation neural networks. However, it is discovered that gradient boosting techniques and random forest models both perform better than SVRs (Ho *et al.*, 2021). The best results are produced by the RIPPER method (repeated incremental pruning to produce error reduction), which is compared to naive Bayesian algorithms, C4.5 and AdaBoost. A comparative study of fuzzy inference systems, neural networks and fuzzy least-squares demonstrates the potential advantages of fuzzy-based techniques. The effectiveness of different data processing techniques is demonstrated by a research that compares neural networks with single/double exponential smoothing, group data processing and back-propagation neural networks (Li *et al.*, 2017). The comparison of neural networks' convolution and long short-term memory versions, among many others, shows that the latter performs better than the former. Furthermore, it is shown that long short-term memory neural networks perform better than SVRs, back-propagation neural networks and evolution neural networks (Liu and Liu, 2019). Another study shows that SVRs are not as effective as long short-term memory dense neural networks. SVRs outperform linear regressions, boosting, decision trees and random forests in studies comparing machine learning techniques with conventional econometric methods, and XGBoost outperforms linear regressions, LASSO (least absolute shrinkage and selection operator), Ridge, random forests, bagging and boosting (Yan and Zong, 2020). However, random forests are shown to perform better than nearest neighbors, linear regressions, the XGBRegressor, CatBoost, AdaBoost, linear LASSO and linear Ridge (Rico-Juan and de La Paz, 2021). Furthermore, it is discovered that neural networks (Peterson and Flanagan, 2009; Abidoye and Chan, 2018), random forests, boosting, Ridge, LASSO and bagging (Embaye *et al.*, 2021) all outperform linear regressions in terms of prediction accuracy. Random walk and Bayesian (vector) autoregression models are proven to perform worse than SVRs based on ensemble empirical mode decomposition (Plakandaras *et al.*, 2015). These benefits are also evident when contrasting AR integrated moving average models with multilayer perceptron neural

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networks. Furthermore, studies show that when compared to deep learning neural networks, time-series econometric methodologies and a variety of other machine learning methods, there is no obvious benefit or drawback (Milunovich, 2020). Empirical results on different machine learning algorithms to estimate real estate values are, predictably, equivocal. Certain real estate price time series under consideration may have an impact on the machine learning techniques used by various researchers. Research similar to the ones listed here often point out benefits of machine learning techniques over conventional econometric ones. Our work is one of several that focuses on a single machine learning technique (Lam *et al.*, 2008; Azadeh *et al.*, 2014; Ma *et al.*, 2015; Morano *et al.*, 2015; Abidoye and Chan, 2017; Ćetković *et al.*, 2018; Li *et al.*, 2018; Kang *et al.*, 2020; Yasnitsky *et al.*, 2021; Wang *et al.*, 2014; Li *et al.*, 2020; Gu *et al.*, 2011; Rafiei and Adeli, 2016; Chen *et al.*, 2017) using Gaussian process regressions to anticipate real estate price time series. It has been suggested in a number of different research that integrating several models is a workable way to increase forecast accuracy. These include combining neural networks with multiple linear regressions, averaging dynamic models (Wei and Cao, 2017) and using neural networks in conjunction with case-based reasoning. These research use machine learning techniques to try to make up for any linear econometric underestimations of real estate pricing nonlinearities.

Various machine learning techniques have shown encouraging accuracy for real estate price projections in the literature. Based upon diverse evidence from empirical research, the measurement of the mean absolute percentage forecast errors could be below 1% (Liu and Wu, 2020; Ma *et al.*, 2015; Pai and Wang, 2020; Liu and Liu, 2019; Ho *et al.*, 2021), stretching from 1% to 2% (Liu and Wu, 2020; Ma *et al.*, 2015; Pai and Wang, 2020; Rico-Juan and de La Paz, 2021), stretching from 2% to 3% (Pai and Wang, 2020; Liu and Liu, 2019; Plakandaras *et al.*, 2015), stretching from 3% to 4% (Liu and Wu, 2020; Wang *et al.*, 2014; Rafiei and Adeli, 2016), stretching from 4% to 5% (Kang *et al.*, 2020; Wang *et al.*, 2014), stretching from 5% to 6% (Liu and Wu, 2020; Kang *et al.*, 2020; Li *et al.*, 2020; Plakandaras *et al.*, 2015), stretching from 6% to 7% (Liu and Wu, 2020; Kang *et al.*, 2020), stretching from 7% to 8% (Pai and Wang, 2020; Kang *et al.*, 2020) and stretching from 8% to 9% (Kang *et al.*, 2020) [2]. More precisely, creating random forecasts (Ho *et al.*, 2021), gradient boosting (Ho *et al.*, 2021), neural networks (Ma *et al.*, 2015), modified Holt's exponential smoothing (Liu and Wu, 2020) and SVRs (Pai and Wang, 2020; Ho *et al.*, 2021) results in the mean absolute percentage forecast error that is less than 1%. Creating neural networks (Ma *et al.*, 2015), XGBoost, SVRs (Pai and Wang, 2020) and modified Holt's exponential smoothing (Liu and Wu, 2020) results in the mean absolute percentage forecast error stretching from 1% to 2%. Creating SVRs (Liu and Liu, 2019; Plakandaras *et al.*, 2015), classification and RT (Pai and Wang, 2020) and neural networks (Liu and Liu, 2019) results in the mean absolute percentage forecast error stretching from 2% to 3%. Creating neural networks, SVRs (Wang *et al.*, 2014), deep restricted Boltzmann machines (Rafiei and Adeli, 2016) and modified Holt's exponential smoothing (Liu and Wu, 2020) results in the mean absolute percentage forecast error stretching from 3% to 4%. Creating genetic algorithms (Kang *et al.*, 2020), neural networks and SVRs (Wang *et al.*, 2014) results in the mean absolute percentage forecast error stretching from 4% to 5%. Creating neural networks (Liu and Wu, 2020), SVRs (Li *et al.*, 2020; Plakandaras *et al.*, 2015) and genetic algorithms (Kang *et al.*, 2020) results in the mean absolute percentage forecast error stretching from 5% to 6%. Creating neural networks (Liu and Wu, 2020) and genetic algorithms (Kang *et al.*, 2020) results in the mean absolute percentage forecast error stretching from 6% to 7%. Creating genetic algorithms (Kang *et al.*, 2020) results in the mean absolute percentage forecast error stretching from 7% to 8%. And creating neural networks (Pai and Wang, 2020) and genetic algorithms

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(Kang *et al.*, 2020) results in the mean absolute percentage forecast error stretching from 8% to 9%. A model may perform well even with significantly larger mean absolute percentage forecast errors because different time-series data have different properties and some are harder to anticipate than others.

Last but not least, it is critical to remember that, like conventional econometric techniques, machine learning methods – including the Gaussian process regression models considered here – used for predictions may run into the overfitting or underfitting problem. Thus, we considered the trade-off between model prediction accuracy and forecast stabilities while developing the final model parameters and estimating pre-owned residential real estate price indices across the ten cities using Gaussian process regressions. The complexity of the model's implementation may increase when comparing machine learning approaches to conventional econometric methodologies. Thus, it could be advisable to recommend simpler model topologies, especially for prediction users who lack experience. The Gaussian process regression models developed here have very basic structures as its foundation, can be implemented with ease and differ just little in running time from several classic econometrics models.

3. Data

The China Real Estate Index System (CREIS) provided the study's data. It is an analytical tool designed to show how the main Chinese cities' real estate markets are doing as well as their growth patterns. The National Real Estate Development Group Corporation, the Development Research Center of the State Council and the Real Estate Association developed the platform in 1994. Experts from the Banking Regulatory Commission, the Real Estate Association, the Development Research Center of the State Council, the Ministry of Land and Resources, the Ministry of Construction and other universities audited CREIS in 1995 and 2005. These days, CREIS releases a variety of real estate price indices each month, such as price indices for villas, retail real estate, office space, rental properties and residential real estate sales price indices for both newly built and existing homes, among many other categories. From a platform, this system has grown to encompass most of the Chinese real estate markets (Yang *et al.*, 2013). In this work, we primarily use pre-owned residential real estate price indices to investigate forecasting problems.

The following ten major Chinese cities are included in the pre-owned residential real estate price indices gathered from CREIS: Chengdu, Wuhan, Nanjing, Hangzhou, Guangzhou, Shenzhen, Chongqing, Tianjing, Shanghai and Beijing. Data is gathered by phone, field and online surveys, according to CREIS. Every pre-owned residential property, excluding villas, in a given city that is sold in a given month is included in the samples used to create the price index. With an index value of 1,000, the base-period price index is based on the price of pre-owned residential real estate in Beijing in 12M2004. Pre-owned residential real estate price indices for different places and months are built by normalizing against this base-period price. The following is how CREIS calculates its pre-owned residential real estate price index for a given city: $I_t^{'} = \frac{\sum P_i^t A_i^{t-1}}{\sum P_i^{t-1} A_i^{t-1}} \cdot I_{t-1}^{'}$, where $I_t^{'}$ and $I_{t-1}^{'}$ reflect price indices associated with time t and time $t-1$, respectively, A_i^{t-1} expresses Project i's total area of construction corresponding to time $t-1$, and P_i^t and P_i^{t-1} signify average prices of Project i at time t and time $t-1$, respectively. It is crucial to emphasize that the only data examined in the current inquiry are the pre-owned residential real estate price indices for each of the ten cities. No more CREIS platform data is available to us. The monthly data used in this research spans the period 3M2012–5M2020.

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The ten price indices, their first differences, quantile–quantile plots and distributional plots using histograms and kernel estimations are all visualized in Figure 1. Table 1 also summarizes the ten price indices as well as their first differences. Based on the p-values of the Jarque–Bera test displayed in Table 1, none of the ten price indices exhibits a normal distribution at the 5% significance level. Non-normality is probably not startling for a wide range of financial and economic time-series data types. The price indices of Beijing, Shanghai and Shenzhen are left-skewed and those of other seven cities are right-skewed. For each of the ten cities, the platykurtic price index is determined.

Expressions of nonlinear characteristics at higher moments have previously been widely reported in the domains of finance and economics throughout a broad spectrum of time-series data (Yang *et al.*, 2008; Wang and Yang, 2010; Yang *et al.*, 2010). We use the Brock–Dechert–Scheinkman (BDS) (Brock *et al.*, 1996) test to look for any possible nonlinear patterns in the ten time-series data sets of pre-owned residential housing prices. We apply the BDS test using 2–10 as the embedding dimensions and 0.5, 1.0, 1.5, 2.0, 2.5 and 3.0 multiplied by the standard deviation of a given price index time series as the ϵ's distance, which are used to evaluate the closeness of various data points. We find that all of the test's resulting p-values are almost zero. These results imply that there are nonlinearities in each of the ten price indices. In light of these details, the goal of this study is to use Gaussian process regressions to forecast the ten non-normal and nonlinear pre-owned residential housing price indices.

4. Method

The forecasting method being studied in this work is the Gaussian process regression, a type of probabilistic kernel model that has been shown to be successful at predicting a range of nonlinear patterns in a number of scientific areas (Ou and Wang, 2011; Zhang and Xu, 2020a; Jin and Xu, 2024d; Zhang and Xu, 2020b; Han and Zhang, 2015). The training data having an uncertain distribution for the model's representation are denoted by $\{(x_i, y_i); i = 1, 2, \ldots, T\}$, the predictors in d-dimensions are shown by $x_i \in \mathbb{R}^d$, and the target is represented by $y_i \in \mathbb{R}$. Nine lagged price indices are used as predictors to forecast

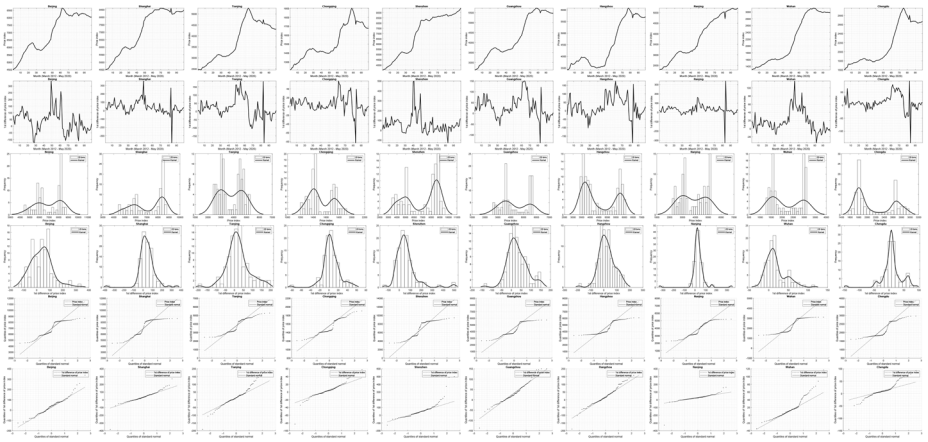

Figure 1.
Pre-owned residential housing price indices of ten major cities in China for the period 3M2012–5M2020

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Table 1. Summary statistics of pre-owned residential housing price indices of ten major cities in China for the period 3M2012–5M2020

City	Series	Min.	1st percentile	5th percentile	Mean	Median	SD	95th percentile	99th percentile	Max.	Skewness	Kurtosis	Jarque–Bera (%)
Beijing	Price	4480.0000	4518.7100	4726.2000	6955.2121	7127.0000	1285.0890	8446.6500	8634.1800	8643.0000	−0.2701	1.6104	1.9119
	First difference	−125.0000	−120.6800	−81.6000	36.4796	40.5000	81.9017	183.8000	300.0800	338.0000	0.6792	4.3601	0.6633
Shanghai	Price	4477.0000	4493.1700	4647.4500	7085.0202	7819.0000	1424.2927	8578.6500	8605.5100	8606.0000	−0.3840	1.5493	1.2853
	First difference	−385.0000	−238.1200	−46.6000	40.5714	33.0000	79.7011	166.0000	313.1600	353.0000	−0.5495	12.3049	<0.1000
Tianjing	Price	2447.0000	2451.4100	2487.9500	3785.1919	3722.0000	889.7638	5087.2000	5250.3200	5266.0000	0.0234	1.4446	1.6156
	First difference	−192.0000	−150.7200	−75.4000	18.9592	13.5000	62.9239	142.8000	184.5600	186.0000	0.2451	4.1967	3.3211
Chongqing	Price	1210.0000	1212.9400	1261.4000	1521.0404	1441.0000	188.3386	1835.8500	1896.6300	1903.0000	0.3541	1.8370	2.7022
	First difference	−83.0000	−59.4800	−22.0000	4.7755	3.5000	18.3407	38.2000	54.1600	58.0000	−0.5223	8.1308	<0.1000
Shenzhen	Price	3524.0000	3527.9200	3692.9500	6927.5455	8235.0000	1976.2603	9084.9500	9372.1600	9429.0000	−0.4152	1.4998	1.0689
	First difference	−128.0000	−91.0400	−29.8000	60.2551	46.0000	94.4175	211.8000	495.2400	501.0000	2.4024	11.3346	<0.1000
Guangzhou	Price	2622.0000	2626.4100	2747.0000	4287.0707	3855.0000	1029.9256	5662.8000	5724.5900	5729.0000	0.1615	1.5242	1.8174
	First difference	−137.0000	−99.0800	−34.6000	29.1224	20.0000	49.0815	130.2000	150.0000	150.0000	0.2904	3.9554	5.5373
Hangzhou	Price	3412.0000	3423.7600	3504.1500	4516.0808	3958.0000	964.5214	5919.4500	6067.6500	6075.0000	0.4022	1.3989	0.8820
	First difference	−190.0000	−147.7600	−57.0000	23.1327	14.5000	60.8186	126.8000	166.8800	176.0000	0.0349	3.7869	19.0631
Nanjing	Price	2278.0000	2289.7600	2385.7000	3827.3737	3716.0000	986.0573	5182.6000	5209.1600	5217.0000	0.0338	1.4962	1.8428
	First difference	−328.0000	−185.4400	−21.4000	29.9898	25.0000	58.0335	93.4000	254.0400	339.0000	−0.6255	24.1061	<0.1000
Wuhan	Price	1292.0000	1300.8200	1349.8500	2188.9091	1904.0000	690.1899	3096.0000	3111.0600	3114.0000	0.1864	1.2693	0.9304
	First difference	−35.0000	−27.3200	−13.6000	17.1939	12.0000	25.7789	67.4000	103.0800	117.0000	1.1776	4.6712	<0.1000
Chengdu	Price	1599.0000	1603.9000	1688.5500	2155.8586	1879.0000	441.8700	2874.4000	2956.6100	2962.0000	0.5051	1.5454	0.9291
	First difference	−145.0000	−98.9200	−39.0000	10.9184	9.0000	31.5185	67.0000	86.2800	93.0000	−0.7713	8.3599	<0.1000

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the pre-owned housing price indices for each city. For instance, nine price indices from the nine months prior will be used as predictors to anticipate the price index for the tenth month.

Let $y = x^T\beta + \varepsilon$ express a linear regression, where $\varepsilon \sim N(0, \sigma^2)$ reflects the error term. In contrast, the target variable in Gaussian process regressions is defined using latent variables and explicit basis functions (Zhang and Xu, 2020c). It is possible to describe the basis function using b and the latent variables from the Gaussian process using $l(x_i)$ in a way that satisfies the joint Gaussian distribution condition. The target's smoothness will be represented by the latent variables' covariance function, and the basis function's objective is to project different predictors onto the feature space (Zhang and Xu, 2020d).

Two metrics that are commonly used to characterize a Gaussian process (*GP*) are the covariance and mean (Zhang and Xu, 2020e). We will use $m(x) = E(l(x))$ to represent the mean, and $k(x, x') = \mathrm{Cov}[l(x), l(x')]$ to express the covariance. Next, we will use $y = b(x)^T\beta + l(x)$ to represent the Gaussian process regression, where $l(x) \sim GP(0, k(x, x'))$ and $b(x) \in \mathbb{R}^p$. We will parameterize $k(x, x'|\boldsymbol{\theta})$ via θ, a hyper-parameter. The following variables will often be estimated while training a Gaussian process regression using a particular technique: θ, σ^2 and β. In addition, we will specify the kernels (represented as k's) and basis functions (represented as b's) that will be used in the training of the model (Zhang and Xu, 2020f). Isotropic kernels and nonisotropic kernels – also known as automatic relevance determination kernels – are the two types of kernels that are taken into consideration in this study. Five different kernels are used to analyze both isotropic and nonisotropic kernels (Zhang and Xu, 2021a; Zhang and Xu, 2020g). The specifications for each of the kernels under consideration are provided by equations (1)–(10). The characteristic length scale of isotropic kernels is denoted by σ_l, the scale-mixture parameter is indicated by $\alpha > 0$, the signal standard deviation is indicated by σ_f and $r = \sqrt{(x_i - x_j)'(x_i - x_j)}$. $\theta = (\theta_1, \theta_2) = (\log\sigma_l, \log\sigma_f)$ will be used to accomplish the positiveness of σ_l and σ_f (Zhang and Xu, 2021b; Xu *et al.*, 2022). For nonisotropic kernels, each predictor will have a unique length scale, denoted by σ_m $(m = 1, 2, \ldots, d)$. Accordingly, θ is reflected through $\theta = (\theta_1, \theta_2, \ldots, \theta_d, \theta_{d+1}) = (\log\sigma_1, \log\sigma_2, \ldots, \log\sigma_d, \log\sigma_f)$ (Zhang and Xu, 2021c; Zhang and Xu, 2021d):

$$\text{Isotropic Exponential}: k(x_i, x_j|\theta) = \sigma_f^2 e^{-\frac{r}{\sigma_l}} \tag{1}$$

$$\text{Isotropic Squared Exponential}: k(x_i, x_j|\theta) = \sigma_f^2 e^{-\frac{1}{2}\frac{(x_i - x_j)^T(x_i - x_j)}{\sigma_l^2}} \tag{2}$$

$$\text{Isotropic Matern } 5/2: k(x_i, x_j|\theta) = \sigma_f^2\left(1 + \frac{\sqrt{5}r}{\sigma_l} + \frac{5r^2}{3\sigma_l^2}\right)e^{-\frac{\sqrt{5}r}{\sigma_l}} \tag{3}$$

$$\text{Isotropic Rational Quadratic}: k(x_i, x_j|\theta) = \sigma_f^2\left(1 + \frac{r^2}{2\alpha\sigma_l^2}\right)^{-\alpha} \tag{4}$$

$$\text{Isotropic Matern } 3/2: k(x_i, x_j|\theta) = \sigma_f^2\left(1 + \frac{\sqrt{3}r}{\sigma_l}\right)e^{-\frac{\sqrt{3}r}{\sigma_l}} \tag{5}$$

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$$\text{Nonisotropic Exponential}: k(x_i, x_j|\theta) = \sigma_f^2 e^{-\sqrt{\sum_{m=1}^{d} \frac{(x_{im} - x_{jm})^2}{\sigma_m^2}}} \tag{6}$$

$$\text{Nonisotropic Squared Exponential}: k(x_i, x_j|\theta) = \sigma_f^2 e^{-\frac{1}{2}\sum_{m=1}^{d} \frac{(x_{im} - x_{jm})^2}{\sigma_m^2}} \tag{7}$$

$$\text{Nonisotropic Matern } 5/2: k(x_i, x_j|\theta)$$

$$= \sigma_f^2 \left(1 + \sqrt{5\sum_{m=1}^{d} \frac{(x_{im} - x_{jm})^2}{\sigma_m^2}} + \frac{5}{3}\sum_{m=1}^{d} \frac{(x_{im} - x_{jm})^2}{\sigma_m^2}\right) e^{-\sqrt{5\sum_{m=1}^{d} \frac{(x_{im} - x_{jm})^2}{\sigma_m^2}}} \tag{8}$$

$$\text{Nonisotropic Rational Quadratic}: k(x_i, x_j|\theta) = \sigma_f^2 \left(1 + \frac{1}{2\alpha}\sum_{m=1}^{d} \frac{(x_{im} - x_{jm})^2}{\sigma_m^2}\right)^{-\alpha} \tag{9}$$

$$\text{Nonisotropic Matern } 3/2: k(x_i, x_j|\theta) = \sigma_f^2 \left(1 + \sqrt{3\sum_{m=1}^{d} \frac{(x_{im} - x_{jm})^2}{\sigma_m^2}}\right) e^{-\sqrt{3\sum_{m=1}^{d} \frac{(x_{im} - x_{jm})^2}{\sigma_m^2}}} \tag{10}$$

This study considers four distinct basis functions, as described in equations (11)–(14), in a manner akin to the consideration of different kernels. In these equations: $X = (x_1, \ x_2, \ \ldots, x_n)'$, $X^2 = \begin{pmatrix} x_{11}^2 & x_{12}^2 & \cdots & x_{1d}^2 \\ x_{21}^2 & x_{22}^2 & \cdots & x_{2d}^2 \\ \vdots & \vdots & \vdots & \vdots \\ x_{T1}^2 & x_{T2}^2 & \cdots & x_{Td}^2 \end{pmatrix}$, and $B = (b(x_1), \ b(x_2), \ \ldots, \ b(x_n))'$.

$$\text{Empty}: B = \textit{Empty Matrix} \tag{11}$$

$$\text{Constant}: B = I_{n\times 1} \tag{12}$$

$$\text{Linear}: B = [1, X] \tag{13}$$

$$\text{Pure Quadratic}: B = [1, X, X^2] \tag{14}$$

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The model parameters are estimated using tenfold cross-validation and Bayesian optimization, which is based on the expected improvement per second plus (*EIPSP*) approach (Zhang and Xu, 2021e; Zhang and Xu, 2021f). Let $f(x)$ be the expression for a *GP* model. The Bayesian method chooses N_s randomly picked data points of x_i's inside variable boundaries to assess corresponding $y_i = f(x_i)$ (Zhang and Xu, 2022a). In this case, $N_s = 4$ data points were used to make preliminary assessments (Zhang and Xu, 2022a). When evaluation errors are encountered, the algorithm will continue to acquire data until it reaches N_s successful evaluation cases (Zhang and Xu, 2021g). Then, as seen below, the algorithm's first and second phases are repeated. The posterior distribution over $Q(f|x_i, y_i \text{ for } i = 1, \ldots, T)$ will be generated starting with the updating of $f(x)$. Selecting a new data point (x) to ascertain the acquisition function's ($a(x)$) reduction objective will be the second stage. There will be a maximum of 100 iterations (Zhang and Xu, 2021h). The use of $a(x)$ is to assess the goodness of x with respect to Q. Expected improvement acquisition functions assess expected quantities of improvements to the objective function, as opposed to values that would raise the objective function (Zhang and Xu, 2021i). We will let x_{best} represent the data point at which the lowest posterior mean is attained, and $\mu_Q(x_{best})$ indicate the corresponding numerical value of the lowest mean. $EI(x, Q) = E_Q\left[\max\left(0, \mu_Q(x_{\text{best}}) - f(x)\right)\right]$ may be used to indicate the expected improvement (*EI*). By applying the time-weighting method on the acquisition function, the Bayesian technique can provide greater benefits per unit of time, as the location can affect how long it takes to evaluate the target (Zhang and Xu, 2022b). It is feasible to maintain an additional Bayesian model of the time required to assess the goal as a function of x throughout optimization procedures. Considering this, we may write $\text{EIPS}(x) = \frac{EI_Q(x)}{\mu_S(x)}$ to describe the *EI* per second (*EIPS*) of the acquisition function, where $\mu_S(x)$ denotes the posterior mean related to this additional timing *GP* model. The following modifications to the acquisition function's behavior might be made to prevent it from overusing a particular region and from preventing a local minimum of the goal (Zhang and Xu, 2021j). Let $\sigma_F(x)$ represent the standard deviation of the posterior objective corresponding to x, and let σ_{PN} represent the posterior standard deviation of the additive noises satisfying the formula $\sigma_Q^2(x) = \sigma_F^2(x) + \sigma_{PN}^2$. We will use $t_{\sigma_{PN}} > 0$ to describe the exploration ratio. The acquisition function using the *EIPSP* algorithm checks if the next data point, x, satisfies the condition $\sigma_F(x) < t_{\sigma_{PN}}\sigma_{PN}$ after each iteration. If this condition is satisfied, θ will be multiplied by the number of repetitions, and x will be considered overexploiting, and the kernel function will be adjusted accordingly (Bull, 2011). For data points between observations, the *EIPSP* approach modification essentially increases σ_Q (Xu and Zhang, 2022k). The newly fitted kernel will then be used to create a new data point. In the event that the new data point is found to be overly exploitative, θ will be raised ten times in the next trials. To get a data point, x, that is not deemed overexploited, this approach will be used up to five times (Zhang and Xu, 2021k). The *EIPSP* algorithm will then accept the changed x as the subsequent exploration ratio. The algorithm finds a balance between examining fresh data points and previously examined neighboring data points to produce an overall answer that is more accurate (Zhang and Xu, 2021l).

We will perform Bayesian optimization operations over σ, basis functions, kernels and standardization status of predictors (Zhang and Xu, 2022c). The RRMSE, which enables comparisons of various prediction outcomes across several models or goals, will be used to assess forecast performance (Despotovic *et al.*, 2016). The RRMSE may be written like this:

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$RRMSE = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i^{obs} - y_i^{for}\right)^2}}{\frac{1}{n}\sum_{i=1}^{n} y_i^{obs}}$, where n denotes the number of observations used for performance evaluations, y^{for} expresses the target's predicted numerical value and y^{obs} indicates the observed numerical value of the target variable. The mean absolute error (MAE) and root mean square error (RMSE), whose units match the target variable and whose magnitude is tied to the target variable, are two additional performance measures that are used to evaluate prediction accuracy. One way to represent the *RMSE* is: $RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i^{obs} - y_i^{for}\right)^2}$. And one way to express the *MAE* is: $MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i^{obs} - y_i^{for}|$. Finally, we additionally take into account the correlation coefficient (*CC*) for performance evaluation, which is written as $CC = \frac{\sum_{i=1}^{n}\left(y_i^{\text{obs}} - \overline{y^{\text{obs}}}\right)\left(y_i^{\text{for}} - \overline{y^{\text{for}}}\right)}{\sqrt{\sum_{i=1}^{n}\left(y_i^{\text{obs}} - \overline{y^{\text{obs}}}\right)^2 \sum_{i=1}^{n}\left(y_i^{\text{for}} - \overline{y^{\text{for}}}\right)^2}}$, where $\overline{y^{\text{obs}}}$ and $\overline{y^{\text{for}}}$ express averages.

5. Result

Data from pre-owned residential housing price indices for each city are used for model performance testing for projections one month in advance from 6M2019 to 5M2020, as well as for model training from 3M2012 to 5M2019. The results of *EIPSP* optimizations based on training data for all price indices are displayed in Figure 2. These findings suggest that (a) the isotropic exponential kernel [equation (1)], empty basis function [equation (11)] and standardized predictors are selected for the price index of Beijing, (b) the isotropic exponential kernel [equation (1)], constant basis function [equation (12)] and standardized predictors are selected for the price index of Shanghai, (c) the isotropic exponential kernel [equation (1)], empty basis function [equation (11)] and nonstandardized predictors are selected for the price index of Tianjing, (d) the nonisotropic Matern 3/2 kernel [equation (10)], constant basis function [equation (12)] and nonstandardized predictors are selected for the price index of Chongqing, (e) the isotropic Matern 5/2 kernel [equation (3)], empty basis function [equation (11)] and standardized predictors are selected for the price index of Shenzhen, (f) the isotropic Matern 5/2 kernel [equation (3)], linear basis function [equation (13)] and standardized predictors are selected for the price index of Guangzhou, (g) the isotropic exponential kernel [equation (1)], empty basis function [equation (11)] and

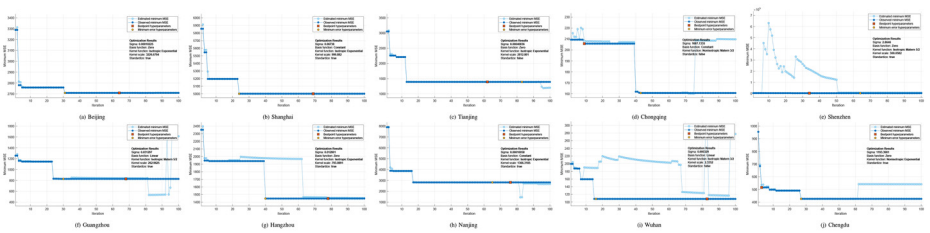

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Figure 2. EIPSP-based optimization results for monthly pre-owned residential housing price indices

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standardized predictors are selected for the price index of Hangzhou, (h) the isotropic exponential kernel [equation (1)], constant basis function [equation (12)] and standardized predictors are selected for the price index of Nanjing, (i) the isotropic Matern 3/2 kernel [equation (5)], linear basis function [equation (13)] and nonstandardized predictors are selected for the price index of Wuhan, and (j) the nonisotropic exponential kernel [equation (6)], empty basis function [equation (11)] and standardized predictors are selected for the price index of Chengdu. The results of parameter estimates for the ten GPR models developed using the tenfold cross-validation for each city's pre-owned residential housing price index are displayed in Table 2. The initials "CV1," "CV2," . . ., and "CV10" are used to signify these parameter estimation results, where 'CV" refers to "cross-validation."

The ten GPR models developed for each city's pre-owned residential housing price index, Models "CV1," "CV2," . . ., and "CV10," are shown in Table 2 and are used to forecast the price index's numerical values for the testing period of 6M2019–5M2020. Therefore, there will be ten anticipated values for each price index for each month of the testing period. The average of the ten estimates is used to get the final price index estimate for that specific month. This method effectively eliminates any potential idiosyncratic forecasts made by a certain submodel, which might contribute to future projections that are stable and dependable. The advantages and desired characteristics of equal weighing have been covered in the literature (Costantini *et al.*, 2017). Figure 3 compares the pre-owned residential housing price indices for each city, both predicted and actual. The percentage forecast errors are shown in Figure 4, which is consistent with Figure 3. It is evident that observed price indices and anticipated price indices generally track each other rather closely. For the findings in Figures 3 and 4, additional prediction performance data in terms of the RMSE, RRMSE, MAE and CC are summarized in Table 3. Specifically, the ten price indices have RRMSEs ranging from 0.0458% to 0.3035%. According to earlier studies (Despotovic *et al.*, 2016), model prediction accuracy might be ranked as follows: excellent if RRMSE$<$ 10%, good if 10% $<$RRMSE$<$ 20%, fair if 20% $<$RRMSE$<$ 30%, and poor if RRMSE$\geq$ 30%. These criteria indicate that the GPR models developed here have an excellent level of prediction accuracy.

Figure 5 presents the results of an error autocorrelation study that was carried out to assess the adequacy of the constructed models. The study is carried out for up to 20 lags, with an emphasis on normalized autocorrelations. These findings validate the general validity of the models and rule out any blatant autocorrelations. It may be noteworthy, nonetheless, that including the AR conditional heteroscedasticity effect into a prediction model can improve its performance, even in the face of contradictory empirical evidence [e.g. 183, 184].

We also analyze the models' performance for a more recent time period. Specifically, we apply the models shown in Table 2 on pre-owned residential housing prices across the ten cities for the time period of February 2023–January 2024. Recalling that the models use nine lagged series values as predictors, price forecasts are generated for November 2023, December 2023 and January 2024, which are shown as red dots in Figure 6. When comparing the forecasted and observed prices for each city during these three months, as depicted in Figure 6, it could be seen that the forecasts closely track observed prices. This further suggests the usefulness of the constructed models here.

6. Result analysis of contrast forecast models

So far, the investigation has mostly concentrated on regressions using the Gaussian process. The AR model, Arkansas-generalized autoregressive conditional heteroscedasticity (abbreviated as AR-GARCH) model, SVR model and RT model serve as four alternative models considered here for benchmark purposes. While comparing forecast accuracy of

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Table 2. Ten GPR models' parameter estimations for each city's pre-owned residential housing price index

Estimate	CV1	CV2	CV3	CV4	CV5	CV6	CV7	CV8	CV9	CV10
Parameter	Beijing: isotropic exponential kernel, empty basis function and standardized predictors									
σ	11.34644	10.88397	11.21652	11.25806	11.06121	11.35184	11.23226	11.36280	11.28581	11.29736
σ_l	1478.76297	1549.80495	1511.41873	1487.23894	1469.36263	1527.77996	1584.82872	1415.07405	1456.23803	1431.98451
σ_f	6996.28789	7058.99367	6964.91820	6985.32766	6985.61621	6995.05097	6992.20871	6984.68920	6997.99299	6986.10874
Parameter	Shanghai: isotropic exponential kernel, constant basis function and standardized predictors									
σ	12.60171	12.65667	12.43396	12.46169	12.53081	12.30951	12.44506	12.68133	12.61629	12.73437
β	6716.31619	6722.91918	6726.56064	6716.64362	6722.22576	6767.73831	6709.08306	6728.25743	6718.44492	6741.19175
σ_l	76.71800	79.93348	105.42119	76.96624	77.65031	90.95637	83.80150	78.51796	76.06861	88.07812
σ_f	1734.47766	1744.49917	1771.82939	1731.23332	1737.49918	1752.39015	1736.88704	1739.06683	1735.42166	1772.84554
Parameter	Tianjing: isotropic exponential kernel, empty basis function and nonstandardized predictors									
σ	8.12223	8.37052	8.41959	8.14719	8.47991	8.33426	8.26314	8.16722	8.36124	8.29159
σ_l	917896.68400	901786.73526	1118250.47428	907284.88983	897970.04107	906877.37576	905286.14179	903439.65243	909402.33100	968360.07350
σ_f	4059.19284	4054.76853	4068.70065	4073.58017	4057.06349	4054.87394	4057.33326	4054.47449	4056.23212	4044.77978
Parameter	Chongqing: nonisotropic Matern 3/2 kernel, constant basis function and nonstandardized predictors									
σ	5.56727	4.92674	6.50467	4.28530	5.35656	5.54645	5.15887	4.11237	5.12630	4.26242
β	1586.83743	1580.41587	1540.93302	1580.81475	1573.37302	1570.65790	1576.28943	1592.85835	1584.82359	1591.20074
σ_1	6179.74452	49177165094822.20000	119205550.33128	2149.16216	5046068.48468	549422.88346	86565.88130	8264.27540	3572.51910	350.38719
σ_2	336.74491	309.43910	493.70744	262.24809	327.47981	313.34918	291.59866	1221.48760	303.39510	946.63918
σ_3	22830601.42381	50671823250.42500	346209.78951	22342314.86463	490489.89502	4896876.17514	1057550.17589	294.88070	1741581.29388	437.97748
σ_4	716.25275	612.09713	11909055.82648	615.48868	846.77062	505.22669	623.48021	1049.59618	582.52378	900.10337
σ_5	786574.36133	35564030.74322	881.36821	328699.26660	755193.56484	702460.44349	2050.46089	1881.80383	2267.40498	292678.05468
σ_6	157283006.40471	401551.22127	208620117.94988	2094464.49521	3422283.88256	1309663.50500	52978396.48421	1463779.80429	2184692.36164	1827.29584
σ_7	434384.33263	2207.38444	91449826.94185	1322.68043	2303.49598	2191.14959	2570.99901	2117640.48482	2859.70570	1208.41138
σ_8	51504378.45013	26348541.91946	97714088.99401	865739.50526	418217.83527	408326.24827	3158058.50326	1193899.85594	505382.64958	88386.51681
σ_9	289.24758	282.27094	423.70755	242.89171	283.75746	288.84678	264.45721	334.87823	283.76100	358.16086
σ_f	171.92764	162.12951	195.82911	151.98030	166.74296	159.13466	159.44168	182.07565	161.32802	152.96275
Parameter	Shenzhen: isotropic Matern 5/2 kernel, empty basis function and standardized predictors									
σ	56.90234	50.35853	43.96349	40.80814	53.80888	53.78936	53.43056	50.46201	54.90458	47.14572
σ_l	137.80039	5.02011	4.27545	4.08478	5.84916	6.07866	7.53360	4.96700	5.58126	4.56867
σ_f	102474.71802	4142.45441	3681.64696	3655.99229	4585.50482	4905.75759	5777.39136	4075.73372	4240.31194	3817.98637
Parameter	Guangzhou: isotropic Matern 5/2 kernel, linear basis function and standardized predictors									
σ	35.68043	34.18927	24.26637	34.11388	35.07319	35.77847	34.52065	34.65384	35.83448	34.24232
β_0	4485.09877	4430.97531	4383.06173	4466.61728	4395.40741	4461.72840	4422.58025	4507.07407	4460.56790	4395.77778
β_1	−22.00451	204.20283	−51.03961	−40.10682	8.34190	−7.28323	−0.69112	−56.28909	−22.18170	−47.39137
β_2	−64.28505	−469.89481	102.95413	−10.60102	−32.16166	−64.93041	−82.15014	16.25216	−70.32153	−25.97377
β_3	143.68304	309.31510	−82.56766	111.21312	75.83371	123.12413	116.38854	118.48221	164.64201	184.78810
β_4	6.24641	−11.41629	95.92353	32.97490	21.68781	18.38982	−21.92609	−42.74598	−14.93738	−41.29688
β_5	−31.28362	9.24901	−47.84468	−54.63476	−120.49049	−47.24500	10.28977	−0.75441	−29.30189	26.58770
β_6	−232.27140	−238.26042	−172.98486	−255.27472	−187.78546	−229.54177	−242.82906	−244.53654	−244.58152	−435.32932
β_7	−11.33771	−7.90012	−19.14608	−42.01500	33.16940	−23.40106	24.85018	−20.72777	−5.30953	140.97066
β_8	−199.90394	−222.54370	−432.65164	−118.83366	−189.02693	−181.22494	−184.85978	−149.52617	−173.64549	−145.39190

(*continued*)

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Table 2.

Estimate	CV1	CV2	CV3	CV4	CV5	CV6	CV7	CV8	CV9	CV10
β_9	1362.03626	1376.23692	1555.03505	1306.75678	1340.30623	1357.39790	1323.55188	1338.44435	1339.00020	1297.67889
σ_l	266648.70619	177047.86936	136094.68457	40617.41750	175184.11308	255379.07297	170519.28893	19638.07055	175799.13066	182805.47349
σ_f	0.00348	0.00579	0.00331	0.00509	0.00599	0.00335	0.00418	0.00448	0.00626	0.00440
Parameter Hangzhou: isotropic exponential kernel, empty basis function and standardized predictors										
σ	9.76159	9.56312	9.55361	9.67115	9.56680	9.50042	9.55300	9.76237	9.50078	9.59281
σ_l	1285.12787	1274.69808	1547.22868	1318.46341	1337.76567	1275.58168	1286.01475	1265.18415	1304.20902	1351.82417
σ_f	4945.14959	4944.55519	4972.09375	4964.27398	4950.56690	4947.07997	4941.23625	4947.08398	4948.16021	4934.15668
Parameter Nanjing: isotropic exponential kernel, constant basis function and standardized predictors										
σ	9.16982	9.00530	9.06427	9.07706	9.04644	9.10117	9.11421	9.30739	9.13766	9.21620
β	3892.37341	3913.27761	3893.10728	3892.44807	3881.41157	3886.87868	3892.75172	3893.17966	3894.86639	3851.13284
σ_l	93.19057	94.38255	93.95045	94.84893	93.50798	93.57927	93.60363	91.29244	110.31117	172.21773
σ_f	1369.19857	1349.25675	1372.45126	1369.86826	1362.07899	1369.59562	1369.08982	1368.53175	1383.41862	1368.59079
Parameter Wuhan: isotropic Matern 3/2 kernel, linear basis function and nonstandardized predictors										
σ	6.66189	6.66678	6.70195	6.59718	6.70471	6.62291	6.63637	6.71766	6.64260	6.73219
β_0	10.78421	12.30100	8.34022	9.92236	8.67804	9.90088	10.04611	4.80144	9.94509	9.61494
β_1	0.06340	−0.01005	0.08518	0.01621	0.10453	0.04171	0.10898	0.10672	0.07988	0.13736
β_2	0.02091	0.09498	−0.03409	0.08375	−0.00834	0.04736	−0.06268	−0.03537	−0.03054	−0.27797
β_3	−0.18118	−0.23945	−0.15890	−0.19974	−0.18776	−0.19678	−0.14400	−0.20213	−0.13627	0.15125
β_4	0.01215	0.22741	0.18200	0.20006	0.20754	0.16922	0.14618	0.19804	0.15205	−0.01339
β_5	0.07525	−0.12087	−0.23858	−0.32256	−0.46272	−0.20585	−0.23221	−0.21823	−0.20137	−0.11539
β_6	−0.38745	−0.25529	−0.14906	−0.02922	0.10892	−0.13963	−0.03333	−0.12452	−0.16991	−0.12346
β_7	0.52049	0.35388	0.44930	0.25959	0.13688	0.28213	0.16985	0.35542	0.28050	0.18125
β_8	−0.73962	−0.63818	−0.76253	−0.54932	−0.35954	−0.54483	−0.47243	−0.84189	−0.50952	−0.34950
β_9	1.61237	1.58255	1.62370	1.53765	1.45720	1.54332	1.51612	1.76027	1.53154	1.40643
σ_l	2.33523	2.33523	2.33523	2.33523	2.33523	2.33523	2.33523	2.33523	2.33523	2.33523
σ_f	11.75326	12.07343	11.78144	12.54297	11.67167	12.49449	11.78219	11.26843	12.26267	10.18165
Parameter Chengdu: nonisotropic exponential kernel, empty basis function and standardized predictors										
σ	4.39657	4.35434	4.36167	4.32293	4.30320	4.26143	4.33931	4.42181	4.36253	4.33344
σ_1	2941.14103	3113.62131	4126.33116	3394.64755	3035.25359	3326.19868	2772.82517	2531.98948	49868618.51215	3165.31256
σ_2	1753725.15362	12531925823.41980	7690572.75182	21414387728.08100	3174699.07872	1578935197.54663	3499884.40862	333960828.84330	10253919.14633	9210839.14545
σ_3	747467.03086	2423070835324.76000	16638.50742	3121288.20940	11177.08409	7834.78238	4377932.31528	1082289924.03462	2043.39925	10300.21350
σ_4	2379755.03833	262120010551.68600	16228.37347	7336002.13208	5164102.63706	43696532.85531	34097968.52278	3138950.57505	1221194.45053	15467880.24444
σ_5	2453.85132	3291.17006	44321565.23677	2436.22905	406210.35107	20672747.73771	32024331.46233	4066.28926	11228642956.61260	2222792.75629
σ_6	2051529.37133	111853876525.18900	4732922.50704	367421 35.35967	904568.47167	2927.63719	505815505.38421	133094027.20823	8086.31028	2820.17721
σ_7	19314.05288	5450.59856	71139689.97648	265471447.04099	2618.58029	6603.49376	2507.52156	20225.60903	469870280.78414	2308567.46543
σ_8	22172970.43821	6546614768.15480	93380908.06771	163324014.66795	12089925.76055	2437685695.62122	1465150.15617	1426404.05317	1611.75648	6046398.86219
σ_9	622.38300	630.41588	892.41631	643.90420	634.75983	651.68949	623.51967	631.20710	766.86019	648.68741
σ_f	2403.72982	2402.25181	2408.48875	2414.05276	2402.35983	2390.71079	2399.17739	2397.75208	2407.90511	2406.04517

Source: Table by authors

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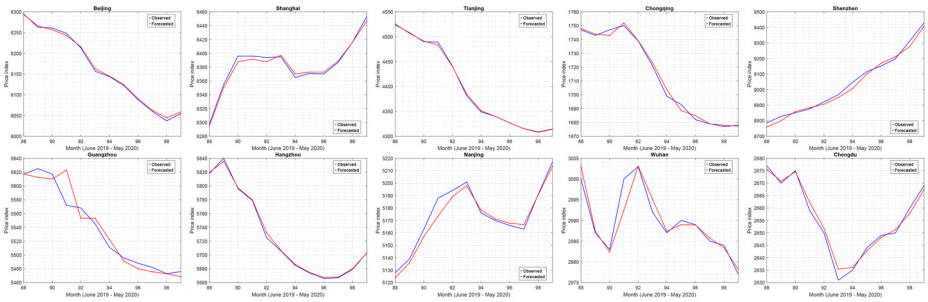
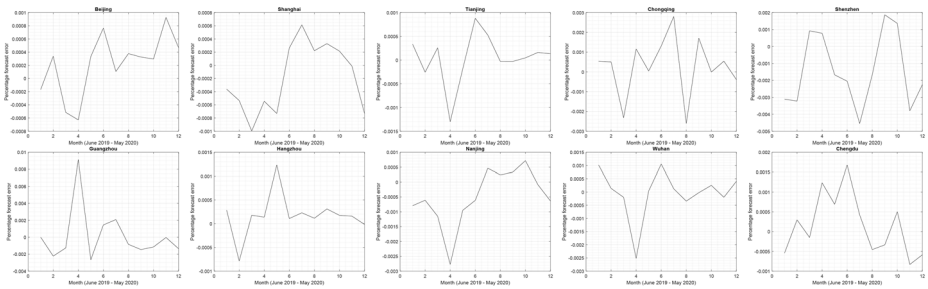
these different models, two measures are adopted: the RRMSE and the modified Diebold–Mariano (MDM) test (Harvey *et al.*, 1997), latter of which evaluates variations related to mean squared errors (MSEs) of forecast results based upon two models. The basis for the MDM test is $d_t = (\text{error}_t^{M_1})^2 - (\text{error}_t^{M_2})^2$, where $\text{error}_t^{M_1}$ indicates the error term derived from model M_1 at time t and $\text{error}_t^{M_2}$ indicates the error term derived from model M_2. M_1 would be one specific benchmark model (AR, Arkansas-GARCH, SVR or RT) and M_2 would be the GPR for the price index of a given city. The way to represent the MDM test statistic is



Source: Figure by authors

Figure 3. The predicted versus observed series plotted for pre-owned residential housing price indices from 6M2019 to 5M2020 during the testing phase



Source: Figure by authors

Figure 4. The percentage projection error plot for pre-owned residential housing price indices from 6M2019 to 5M2020 throughout the testing phase

City	Testing RRMSE (%)	Testing RMSE	Testing MAE	Testing CC (%)
Beijing	0.0492	4.0178	3.5657	99.951
Shanghai	0.0535	4.4844	3.8834	99.314
Tianjing	0.0511	2.2500	1.5302	99.965
Chongqing	0.1498	2.5657	1.9898	99.638
Shenzhen	0.2535	22.9195	20.5673	99.540
Guangzhou	0.3035	16.8132	10.9454	96.031
Hangzhou	0.0458	2.6248	1.7873	99.927
Nanjing	0.1026	5.3085	4.0407	98.151
Wuhan	0.0866	2.5888	1.5776	93.916
Chengdu	0.0761	2.0217	1.7062	99.217

Source: Table by authors

Table 3. GPR models' projected performance for the pre-owned residential housing price indices of ten cities from 6M2019 to 5M2020 during the testing phase

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through $MDM = \left[\frac{T+1-2h+T^{-1}h(h-1)}{T}\right]^{1/2}\left[T^{-1}\left(\gamma_0 + 2\sum_{k=1}^{h-1}\gamma_k\right)\right]^{-1/2}\overline{d}$, where the length of the time period used for contrasting performance is represented by T, the horizon is denoted by h (in our situation, $h = 1$), d_t's average is $\overline{d}$, d_t's variance is $\gamma_0 = T^{-1}\sum_{t=1}^{T}\left(d_t - \overline{d}\right)^2$ and d_t's kth auto-covariance is $\gamma_k = T^{-1}\sum_{t=k+1}^{T}\left(d_t - \overline{d}\right)\left(d_{t-k} - \overline{d}\right)$ for $k = 1, \ldots, h-1$ and $h \geq 2$. The equality of expected forecast performance generated by two distinct models

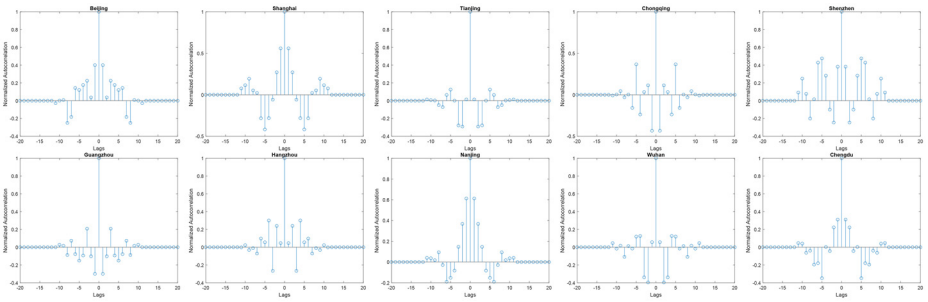


Figure 5. Analysis of error autocorrelations based on GPR models for ten cities' pre-owned residential housing price indices

Source: Figure by authors

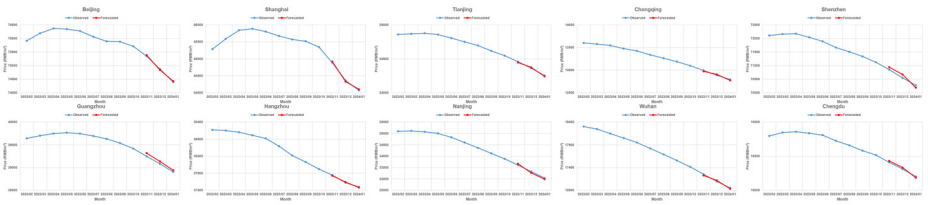
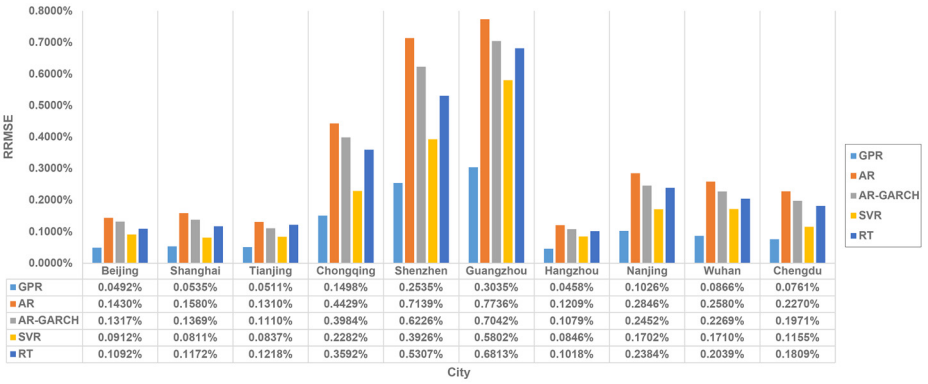


Figure 6. Model applications on pre-owned residential housing prices from February 2023 to January 2024

Source: Figure by authors



	Beijing	Shanghai	Tianjing	Chongqing	Shenzhen	Guangzhou	Hangzhou	Nanjing	Wuhan	Chengdu
GPR	0.0492%	0.0535%	0.0511%	0.1498%	0.2535%	0.3035%	0.0458%	0.1026%	0.0866%	0.0761%
AR	0.1430%	0.1580%	0.1310%	0.4429%	0.7139%	0.7736%	0.1209%	0.2846%	0.2580%	0.2270%
AR-GARCH	0.1317%	0.1369%	0.1110%	0.3984%	0.6226%	0.7042%	0.1079%	0.2452%	0.2269%	0.1971%
SVR	0.0912%	0.0811%	0.0837%	0.2282%	0.3926%	0.5802%	0.0846%	0.1702%	0.1710%	0.1155%
RT	0.1092%	0.1172%	0.1218%	0.3592%	0.5307%	0.6813%	0.1018%	0.2384%	0.2039%	0.1809%

Figure 7. Benchmark analysis in the phase of out-of-sample testing (6M2019–5M2020)

Source: Figure by authors

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is the null hypothesis of the MDM test. The MDM test would come after the t-distribution with $T - 1$ degrees of freedom under the null hypothesis. The AR and AR-GARCH models use the same number of lags as the predictors in the GPR. The GARCH component has the form of GARCH(1,1). The predictors used by the GPR are also used by the SVR and RT models. The classification analysis and regression tree is the method that the RT model uses and the linear ϵ-insensitive version SVR is applied here. In the phase of out-of-sample testing, Figure 7 presents findings of benchmark analysis based upon RRMSEs. It is evident that the GPR has superior accuracy because, for each city's price index, it yields the lowest RRMSE. The GPR is also compared with all four benchmark models in the MDM tests, and the resultant p-values are less than 0.01 for each city's price index. This suggests that, when compared with the four benchmark models under investigation, the GPR generates accuracy that is statistically significantly better.

7. Implication

Forecasts of residential real estate price indices are a significant concern for governments and investors alike. For the purposes of risk management, strategic planning and portfolio allocation and modifications, investors require real estate price projections. For the purpose of assessing the market, developing, implementing and modifying policies – particularly to avoid overheating the market and stimulate the economy when needed – real estate price projections are essential to policymakers. To the best of the authors' knowledge, econometric techniques – particularly time-series approaches where price indices are relevant – are frequently the basis for forecasting and valuation methodologies used by many investors, including those in the public sector. Professional opinions from specialists are also still used. This makes sense since econometric techniques and expert opinions are probably not too difficult to create, use and maintain. They have also been extensively used for many years by forecast users, and many of them may be able to provide a good degree of prediction accuracy. Although some decision-makers still see machine learning designs as unduly complex tools for making forecasts, it may be challenging for some policymakers and investors to take these models into consideration. Nevertheless, it is generally agreed that these models are worth exploring for their potential, especially in light of the growing accessibility of computational capabilities and the realistic basis for potential irregularities in price time-series data. In fact, a number of astute investors and decision-makers have recently shown a growing interest in machine learning techniques for predicting real estate prices. The work being done here carries on the legacy of examining how Gaussian process regressions may be used to tackle forecasting issues for indices of pre-owned residential housing prices. The models developed here could be straightforwardly applied on historical housing price indices of pre-owned residential properties of a city, with the reported model parameters, to generate price index forecasts into the future, thus offering necessary forecast information as part of various decision-making processes. It is worth attention that the present methodology does not take into account other economic factors as potential predictors but focuses on historical information for sources of predictive information. Thus, when using forecast results produced by the models constructed here, one might also combine them with other sources of forecasts. Under this circumstance, appropriate model combinations might need to be established. Overall, the presented methodology here for creating these forecast models for the ten largest Chinese cities, along with their demonstrated high prediction stability and accuracy, indicate that machine learning techniques are worthwhile exploring – possibly for a wider range of real estate types and locations.

8. Conclusion

The current study's focus is on pre-owned residential housing price index estimations for ten significant Chinese cities. We create forecast models using monthly data from 3M2012 to

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5M2020 using the Gaussian process regression technique. We pay special attention to four basis functions, ten kernels and two predictor standardization techniques while developing prediction models with Bayesian optimizations and cross-validation. We find that the developed models yield strong out-of-sample forecasts, with RRMSE ranging from 0.0458% to 0.3035% for the ten price indices during a one-year period from 6M2019 to 5M2020. Market participants and policymakers may find these forecast models useful in improving their comprehension of the pre-owned residential housing sector. Other Bayesian optimization strategies, in addition to the expected improvement per second plus method considered here, might be interesting to explore in future study. It is also possible to expand the forecasting procedure to include a wider range of real estate price indices and more cities/regions.

Notes

1. See ASKCI Consulting Co., Ltd., www.askci.com/
2. None of the earlier studies that are cited here and shown relatively substantial predicted errors have been classified by us.

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